ClipSAT : Cambridge A2 Level Math 9709

Unit 0.01: A2-Level 9709 Paper 3 & Paper 6 Diagnostic Readiness

Name: _____

Q1. Which of the following is the correct expression of dfrac{5x + 1}{x - 1x + 2} in partial fractions?

[ ] A: dfrac{2}{x - 1} + dfrac{3}{x + 2}
[ ] B: dfrac{3}{x - 1} + dfrac{2}{x + 2}
[ ] C: dfrac{2}{x - 1} - dfrac{3}{x + 2}
[ ] D: dfrac{1}{x - 1} + dfrac{4}{x + 2}

Q2. A curve has equation y = ln2x^3 + 1 . What is the gradient of the tangent to the curve at the point where x = 1 ?

[ ] A: 2
[ ] B: 6
[ ] C: dfrac{2}{3}
[ ] D: ln 3

Q3. Find the exact value of the definite integral int_{0}^{ln 2} mathrm{e}^{2x} , mathrm{d}x.

[ ] A: dfrac{3}{2}
[ ] B: 3
[ ] C: dfrac{1}{2}
[ ] D: dfrac{7}{2}

Q4. The rational function f is defined for all real x notin {-1, 2} by fx = frac{x^3 + kx + 7}{x+1x-2^2} where k is a constant. The partial fraction decomposition of fx takes the form A + frac{B}{x+1} + frac{C}{x-2} + frac{D}{x-2^2} where A, B, C, and D are real constants. Given that C = 2 , find the value of k .

[ ] A: -3
[ ] B: 1
[ ] C: 3
[ ] D: 24

Q5. Which of the following is the correct expression of dfrac{5x + 1}{x - 1x + 2} in partial fractions?

[ ] A: dfrac{2}{x - 1} + dfrac{3}{x + 2}
[ ] B: dfrac{3}{x - 1} + dfrac{2}{x + 2}
[ ] C: dfrac{2}{x - 1} - dfrac{3}{x + 2}
[ ] D: dfrac{1}{x - 1} + dfrac{4}{x + 2}

Q6. The complex number z satisfies the equation z2 - mathrm{i} = 7 + 4mathrm{i}. Express z in the form x + mathrm{i}y , where x and y are real.

[ ] A: 2 + 3mathrm{i}
[ ] B: dfrac{18}{5} + dfrac{1}{5}mathrm{i}
[ ] C: 2 - 3mathrm{i}
[ ] D: dfrac{10}{3} + 5mathrm{i}

Q7. A continuous random variable X has probability density function given by fx = frac{3}{8}x^2 for 0 le x le 2 , and fx = 0 otherwise. Find mathrm{P}1 le X le 2 .

[ ] A: dfrac{7}{8}
[ ] B: dfrac{1}{8}
[ ] C: dfrac{3}{4}
[ ] D: dfrac{1}{2}

Q8. A curve has equation y = ln2x^3 + 1 . What is the gradient of the tangent to the curve at the point where x = 1 ?

[ ] A: 2

[ ] B: 6

[ ] C: dfrac{2}{3}

[ ] D: ln 3